

HDV/HDE Booking Tool

Business Requirement Document — v2

Version: 2.0
Date: 2026-04-10
Based on: BRD v1 (original) + Requirements gathered in meetings: 02.04, 03.04, 06.04, 07.04, 08.04, 09.04
Scope: Phase 1 unless noted as Phase 2

Status Legend

Icon	Meaning
	Confirmed — requirement confirmed in meetings, no conflicts
	Partially confirmed / open question — requirement accepted but details TBD
	Phase 2 — deferred to Phase 2; out of Phase 1 scope
	Obsolete / Superseded — cancelled or replaced by another requirement
	New — not in original BRD; identified during requirements gathering
	Conflict — contradiction between BRD and meeting decisions; requires explicit sign-off

1 Introduction

1.1 Purpose

This document defines the business requirements for a booking tool that allows users to create requests for HDV/HDE equipment. Requests may include multiple pieces of equipment, each of which is treated as a separate booking with its own approval workflow.

1.2 Scope

The tool will support bookings for company-owned equipment as well as equipment provided by business partners. It will cover request creation, approval flow, status tracking, reporting, and integration with external systems where applicable.

Phase 1 scope decision (08.04): Booking through the tool is available only for TCO Owned and Long-term rented equipment. On-demand BP (external fleet) is displayed as a catalog/showcase only — no booking workflow for on-demand BP in Phase 1. This is a significant scope reduction vs. original BRD. See Section 6.7 for detail.

2 Business Context

2.1 Current Challenges

- Lack of centralized booking and collaboration between departments on shared resources
- Low transparency on total number of units and spend
- No standardized process or guidelines
- No Process Owner(s)
- Lack of visibility due to manual utilization monitoring

2.2 Business Objectives

- Standardize the request and booking process.
- Ensure automated routing of approval based on equipment fleet ownership.
- Provide transparency and accountability in the booking process.
- Enable scalability for both internal and partner-provided equipment.
- Process ownership by Logistics.

3 Glossary

Request – a container grouping multiple equipment bookings (one per equipment item).

Regular Request – a request created by Requestor in HDV/HDE booking tool.

Service Work Request – a request created in HDV/HDE automatically via API based on Work Order created by Requestor in JDE E1 system.

Booking – individual booking within a request. One booking is linked to one unique equipment item. Treated individually within a request.

Equipment – individual asset (HDV/HDE unit) that can be booked through the system. Each equipment item belongs to a fleet and inherits approval responsibility from the corresponding Fleet Owner.

Fleet – logical group of equipment owned by a Fleet Owner (for internal: Maintenance, Construction, TFM, etc.; for external: business partner's name).

Fleet Owner (FO) – person responsible for approvals within a fleet.

Approver – Fleet Owner and/or Cost Center (CC) owner, who confirms/declines bookings.

 **Mobilization** – the period of preparation/movement of equipment before it arrives at the work site. Booking period starts from mobilization start, not from arrival. *(Мобилизация — период подготовки/перемещения техники до прибытия на объект)*

 Equipment categories (Usage Status):

- **Assigned** – equipment assigned to a specific department; visible to all users, bookable only by authorized users with FO role. *(Закреплённая техника)*
- **Shared without Conditions** – equipment available for general booking. *(Общедоступная без условий)*
- **Shared with Conditions** – equipment available for booking with specific conditions; marked visually (color). *(Общедоступная с условиями)*

 Equipment Ownership Types:

- **TCO Owned** – internally owned by TCO.
- **Long-term rented** – rented long-term; treated as internal fleet for booking purposes.
- **On-demand rented (BP Showcase)** – rented on demand; displayed as catalog/showcase only in Phase 1.

4 Stakeholder & Roles

4.1 Requestor

All users in TCO network will by default have Requestor role to book equipments of the internal fleet (TCO owned + Long-term rented HDV/HDE). Booking of equipments of the external fleet (BP owned HDV/HDE) is available only for users with 'RWA initiator' role in Digital Contractor Timesheet (DCT) tool (Phase 2).

Requestors are allowed to:

1. Create a Regular Request — 
2. Submit the Regular Request — 
3. Prolong booking — 
4. Submit feedback on the booked equipment — 
5. Monitor utilization of the booked equipment for the booking period, if the equipment has tracker(s) — 
6. Terminate booking (only external fleet) —  Phase 2 (external fleet booking is Phase 2)
7.  Close booking by pressing "Close" button (manual close only) —  (*Закрытие брони по кнопке*)
8.  Book Assigned equipment only if authorized by Administrator —  (*Бронирование закреплённой техники — только при авторизации Admin*)

4.2 Service Work Processor

Access provision is managed by Azure Active Directory (AAD) group. Similar to Requestor, however, Service Work Processor works with Service Work Requests, created automatically in HDV/HDE booking tool. — 

4.3 Fleet Owner

For internal Fleet Owners access provision is managed by AAD group. There can be 2+ AAD groups to manage internal Fleet Owners (separate group for each fleet created by Administrator). Fleet Owners are allowed to:

1. Create new equipment under own fleet (allowed only for external Fleet Owners) —  Phase 2 for external
2. Update equipments (only parameters that are allowed for editing by Administrator) — 
3. Delete equipment (soft delete) — 

4. Confirm / decline requests to book equipments under own fleet —
5. Update booking period before confirmation — (conflict with OQ-40: see Section 6.6)
6. Change equipment in booking —
7. Terminate already confirmed bookings (only internal fleet) —
8. Freeze / unfreeze own equipment from booking for a period or without end date —
9. Monitor utilization of own equipment, if the equipment has tracker(s) —
10. Press "Mobilization started" button to fix mobilization start time — (*Кнопка «Мобилизация началась»*)
11. Delegate FO privileges to any TCO employee for a specified period (without IT involvement) — (*Делегирование полномочий FO любому сотруднику TCO*)
12. One user can be an owner of several fleets (member of several AAD groups) —
13. One fleet can have multiple Fleet Owners — Clarification pending (R-48, OQ-30) (*Несколько FO у одного парка*)

4.4 CC Owner (incl. DOA)

There will be no specific AAD group to provide users CC owner's grants. The HDV/HDE tool must identify (via integration with PSWS and DOA) if the user within the tool can conduct actions as CC owner or his DOA. CC owners are allowed to:

1. Confirm / decline external fleet equipment booking request — Phase 2 (*CC Owner approval applies only to external fleet booking, which is Phase 2*)

4.5 Administrator

Administrator role will be provided to process owners – Logistics. Access provision is managed by Azure Active Directory (group). Users with an Administrator role are allowed to:

1. Create/edit internal fleet and link with AAD group —
2. Create/edit external fleet and manage external fleet owners (send invitation, deactivate external user's account) — Phase 2 (external fleet booking)
3. Delete fleet —
4. Create internal fleet equipment —
5. Update equipment's parameters (extended edit) —
6. Delete equipment (soft delete) —
7. Manage 'Request' template —
8. Manage 'Equipment' template —
9. Access to custom reportings —
10. Access to utilization dashboard(s), if the equipment has tracker(s) —

11.  Manage all configurable system parameters (booking horizon, max duration, FO timeout thresholds, etc.) via Admin Panel —  (Управление настраиваемыми параметрами системы через Admin Panel)
12.  Authorize specific users to book Assigned equipment —  (Авторизация пользователей для бронирования закреплённой техники)
13.  Manage delegation settings —  (Управление настройками делегирования)

4.6 Transportation Responsible (Мобилизация/Транспортировка)

A dedicated system role for the team responsible for transporting non-motorized (Unwheeled) equipment (Heavy Ops / Maintenance team). (Ответственный за транспортировку/мобилизацию — Heavy Ops, команда обслуживания)

This role is allowed to:

1.  Receive notification about transportation request for Unwheeled equipment —  (Получение уведомления о запросе на транспортировку)
2.  Confirm or decline the transportation request —  (Подтверждение/отклонение запроса на транспортировку)

5 Entities

5.1 Equipment

All equipments must have a mandatory attribute – fleet. The attribute will affect on which template the equipment will have (depending on internal and external fleet), who can create/edit/delete equipments (fleet owners), approval and processing flow of the equipment booking. — 

Internal fleet equipments have the next parameters:

1. Basic fixed info (TCO ID, Description, Service Zone, Class/Category, License, Serial number, etc.) — 
2. Dynamic (changeable) parameters: Equipment Usage Status, CC/Division/Group/Department, Assigned location, Maintenance BP, Contact number / email, etc.) — 
3. Equipment specifications (picture, Chassis type, Drive type, Loading capacity, Generator / Compressor specifications) — 

4. Trackers' ID if exists (WIALON tracker "Group", IVMS tracker "Registration number", Vega tracker "Device ID", etc) — 

NEW **Equipment Classification (three dimensions):** *(Три измерения классификации техники)*

Dimension	Values
Mobility <i>(Мобильность)</i>	Self-propelled / Not self-propelled (Unwheeled) / Stationary
Ownership <i>(Владение)</i>	TCO Owned / Long-term rented / On-demand rented (BP Showcase)
Usage Status <i>(Статус использования)</i>	Assigned / Shared without Conditions / Shared with Conditions

Note (08.04): Mobility category is NOT displayed in Requestor UI — Requestor sees a flat list of equipment types (forklifts, compressors, cranes, etc.). 

Note (08.04): Stationary HDE is excluded from Requestor search — stored in system for FO use only. 

NEW **Equipment availability display rules:**

- Equipment under maintenance is shown grayed out, not selectable; label "Under maintenance until [Planned date from JDE]"  *(Техника в ремонте — серым, надпись «На ремонте до [дата]»)*
- Repair status and planned completion date are sourced from JDE integration —  (depends on JDE integration)

5.2 Booking

Booking is linked to a single equipment and have booking range (datetime). Several bookings form a single request. — 

NEW **Architecture decision (09.04):** Data model follows Order/OrderLine pattern: one Request contains multiple RequestItems (one per equipment); each RequestItem independently processed by FO and generates a separate Booking.  *(Архитектура: Order / OrderLine — заявка содержит позиции, каждая порождает отдельную бронь)*

5.2.1 Internal Booking

Booking of TCO owned (internal fleet, including Long-term rented) equipment. Internal Fleet equipments require only confirmation from the side of Internal Fleet owner. Fleet owner can apply adjustments to the booking period and/or replace equipment if substitution exists. — 

 **Mobilization (09.04):** Booking period starts from mobilization start, not from arrival at work site. FO presses "Mobilization started" button to fix the actual start in the system. 

5.2.2 External Booking

 **CONFLICT / PHASE CHANGE vs. original BRD:**

Original BRD described a two-step approval flow (CC Owner → External FO) for external fleet equipment. **Meeting decision (08.04): On-demand BP equipment is NOT bookable through the system in Phase 1.** It is displayed as a catalog/showcase only.

- On-demand rented (BP Showcase) — catalog view only; no booking workflow in Phase 1. 
(Phase 1 scope decision)
- Full external booking workflow (CC Owner → External FO → DCT/RWA) —  Phase 2

5.2.3 Booking Snapshot (*Снимок данных техники на момент подтверждения*)

At the moment of booking confirmation, the system saves a snapshot of key equipment attributes (model, license plate, fleet name). This ensures historical readability when equipment data is later updated. —  Preliminary (OQ-37)

5.3 Request

Requests have 2 types: Regular Request, Service Work Request. The type affects on request template, mandatority of filling parameters and approvers' workflow. — 

5.3.1 Regular Request

In a Regular Request Requestor can add 2+ equipments from any fleet; there is no restriction that all equipments must be from the same fleet. Each added equipment will be considered as separate booking and will be confirmed by Fleet Owners separately. — 

The system will prioritize TCO owned (internal fleet) assets as a **business rule** (not technically enforced at current selection model). If FO does not respond within timeout — Requestor is redirected

to the BP catalog. *(Приоритет внутренней техники — бизнес-правило, не технически принудительное)* — ✓

Request fields (08.04):

- **Work Order (JDE)** — mandatory for Maintenance / Railroad / Operations fleets. One Work Order per request. ✓ *(Номер Work Order из JDE — обязательное поле)*
- **Location** (text input) — replaces Work Order field for SCM Logistics fleet. ✓ *(Поле «Локация» для SCM Logistics)*
- **Work Description** — mandatory, ~50 character limit. ✓ *(Описание работ — обязательное поле)*
- **Comments** — optional, no hard limit. ✓ *(Комментарии — опциональное поле)*
- **Priority** (P1–P4) — selected by Requestor. ✓ *(Приоритет P1–P4)*

Request constraints (03.04):

- Maximum booking horizon: 1 month ahead (configurable by Admin). ✓ *(Горизонт бронирования — 1 месяц, настраивается)*
- Maximum single booking duration: 1 week (configurable by Admin). ✓ *(Максимальная длительность — 1 неделя, настраивается)*

 **Form → Bookings split (09.04):** Requestor fills a single form for a Work Order with multiple equipment items. Upon submission, the system **automatically splits** the form into atomic individual bookings — one per equipment item. All bookings from one form are linked via the Work Order field. ✓ *(Одна форма → система автоматически разбивает на атомарные заявки)*

For external fleet equipment added in the Request requestor must provide "Justification" and Cost Center. — ● Phase 2

5.3.2 Service Work Request (within Work Order in JDE E1)

Unlike Regular Request, Service Work Requests are created automatically in the system as a Draft after Work order creation in JDE E1, and submitted by Service Work Processor. Each 'Step' in Work Order will be created as a separate Service Work Request in HDV/HDE booking tool. Service Work Processor assigns equipment(s) and submits the Request. — ⚠ Clarification pending (R-30)

6 Functional Requirements

6.1 User & Role Management

ID	Requirement	Status
FR-001	System supports role-based access control	✓ Confirmed
FR-002	All users in TCO network have access to the tool as a Requestor; to add external fleet equipment, the user must be a member of "RWA initiator" AAD group	● Phase 2 (external fleet booking)
FR-003	Administrators, Service Work Processors are defined by membership in the corresponding AAD groups	✓ Confirmed
FR-004	The system will define if the user has CC Owner or DOA privileges as a result of integration with PSWS and DOA	● Phase 2 (CC Owner role is Phase 2)
 FR-NEW-01	All time-based parameters (FO response timeout, booking horizon, max booking duration) are configurable via Admin Panel and are NOT hardcoded (<i>Все таймауты и временные ограничения настраиваемы через Admin Panel</i>)	✓ Confirmed
 FR-NEW-02	FO can delegate own privileges to any TCO employee for a specified date range (start + end date); delegation is automatically revoked upon expiry; delegation scope is configurable (create requests / approve requests) (<i>Делегирование FO любому сотруднику TCO с настраиваемым объемом и автоотзывом</i>)	✓ Confirmed

6.2 Admin Panel

ID	Requirement	Status
FR-005	Admin can create fleets (internal, external)	✓ Confirmed
FR-006	Admin can link internal fleet with AAD group, so that members of that group can act as Fleet Owners	✓ Confirmed

ID	Requirement	Status
FR-007	Admin can manage owners of created external fleets: send invitation to BP, deactivate existing external accounts	 Phase 2
FR-008	Admin can update existing fleets	 Confirmed
FR-009	Admin can delete existing fleets	 Confirmed
FR-010	Admin can manage template of equipment (internal and external): define parameters, formats, mandatority, editability by owners	 Confirmed
FR-011	Admin can manage template of request	 Confirmed
FR-012	Admin can manage template of booking	 Confirmed
FR-013	Admin can create equipments of internal fleet	 Confirmed
FR-014	Admin can edit (extended edit) equipments	 Confirmed
FR-015	Admin can delete equipments	 Confirmed
FR-016	Admin has access to all dashboards and reports	 Confirmed
 FR-NEW-03	Admin can configure all system time parameters: FO response timeout (default 24h reminder / 48h escalation), max booking horizon, max booking duration (<i>Настройка временных параметров системы через Admin Panel</i>)	 Confirmed
 FR-NEW-04	Admin can authorize specific users (by employee ID or role) to book Assigned equipment and configure justification requirement (<i>Авторизация пользователей для бронирования Assigned техники</i>)	 Confirmed

6.3 Equipment Management

ID	Requirement	Status
FR-017	External fleet owner can create equipments if fills all mandatory parameters	 Phase 2
FR-018	Fleet owner must indicate 'Fleet' of the equipment; system allows selecting only fleets for which the user is a FO	 Confirmed
FR-019	Fleet owner can edit (allowed by Admin) equipments under his fleet; one user can be an owner of several fleets	 Confirmed
FR-020	Fleet owner can delete equipments under his fleet	 Confirmed
FR-021	Fleet owner can freeze / unfreeze own equipments for a period of time or without end date	 Confirmed
FR-022	Requestor / Service Work Processor can submit feedback on equipment with confirmed booking, starting from the booking start datetime	 Confirmed
 FR-NEW-05	FO can upload photos for equipment; Requestor sees equipment photo in the request form (<i>Загрузка фото техники FO; отображение фото в заявке для Requestor</i>)	 Preliminary
 FR-NEW-06	Equipment repair status (maintenance in progress / completion date) is sourced from JDE integration and displayed in Requestor search results (<i>Статус ремонта из JDE — отображается в поиске</i>)	 Pending (OQ — JDE integration detail)
 FR-NEW-07	Stationary HDE equipment is excluded from Requestor search results; stored in system for FO use only (<i>Stationary HDE исключена из поиска Requestor-a</i>)	 Confirmed
 FR-NEW-08	Assigned equipment is visible to all users in search, but can only be booked by users authorized by Admin; "Justification" field is mandatory for Assigned equipment booking (<i>Assigned техника видна всем, бронировать только авторизованным, обязательный Justification</i>)	 Confirmed
 FR-NEW-	Shared with Conditions equipment is displayed with a visual marker (color coding) (<i>Shared with Conditions отображается с</i>	 Confirmed

ID	Requirement	Status
09	<i>цветовым маркером)</i>	

6.4 Request Management

ID	Requirement	Status
FR-023	Requestor can create a Request	✓ Confirmed
FR-024	Request has a unique identifier and metadata (date, requester, status, etc.)	✓ Confirmed
FR-025	Requester can view request details and status	✓ Confirmed
FR-026	Request can have status: Draft, Submitted, In Progress, Completed	⚠ Partially — aggregated statuses TBD (OQ-36)
FR-027	Requester can edit or cancel a draft request before submission	✓ Confirmed
FR-028	Requests created by system automatically (Service Work Requests) will have 'Draft' status before submission by Service Work Processor	✓ Confirmed
FR-029	Only users with role Service Work Processor can manage (submit) Service Work Requests	✓ Confirmed
FR-030	System supports draft saving (continue later)	✓ Confirmed
 FR-NEW-10	TO-BE: only on-demand booking; weekly schedule management is out of scope (<i>Только фактическое бронирование по запросу; еженедельные графики исключены</i>)	✓ Confirmed
 FR-	Requestor selects a booking priority P1–P4 at the time of request creation (<i>Requestor выбирает приоритет P1–P4 при создании заявки</i>)	✓ Confirmed

ID	Requirement	Status
NEW-11		
 FR-NEW-12	Work Order field (JDE) is mandatory for Maintenance/Railroad/Operations fleets; for SCM Logistics fleet this field is replaced by a "Location" (free text) field; Work Order is one-to-one with a request (<i>Поле Work Order обязательно; для SCM Logistics заменяется на Location</i>)	 Confirmed
 FR-NEW-13	"Work Description" field is mandatory in the request form (~50 character limit) (<i>Поле «Описание работ» — обязательное, ~50 символов</i>)	 Confirmed
 FR-NEW-14	"Comments" field is optional in the request form (no hard character limit) (<i>Поле «Комментарии» — опциональное</i>)	 Confirmed
 FR-NEW-15	Requestor submits a single form for a Work Order with multiple equipment items; the system automatically splits the form into individual bookings (one per equipment item); all bookings are linked via the Work Order field (<i>Одна форма → автоматическое разбиение на атомарные заявки</i>)	 Confirmed (OQ-39)
 FR-NEW-16	Partial confirmation: confirmed items within a request proceed independently from declined items (<i>Частичное подтверждение — одобренные позиции продолжают процесс независимо от отклонённых</i>)	 Confirmed
 FR-NEW-17	Aggregated Request status is calculated automatically based on statuses of its RequestItems (<i>Агрегированный статус заявки вычисляется автоматически из статусов позиций</i>)	 Details TBD (OQ-36)

6.5 Booking Management

ID	Requirement	Status
FR-031	Requestor can add/remove multiple equipment items into a single request; equipment availability in DB is updated	✓ Confirmed
FR-032	The system prioritizes internal fleet equipments	✓ Confirmed (as business rule, not technically enforced)
FR-033	Requestor can add internal fleet equipments with "Equipment Usage Status" in (Shared without conditions, Shared with conditions)	✓ Confirmed
FR-034	The system will suggest external fleet equipments based on a pre-set decision matrix (cost, reliability, availability)	✗ Obsolete — replaced by BP showcase model (FR-NEW-28); no booking through system for on-demand BP in Phase 1
FR-035	Requestor can add external fleet equipment to Request, only if RWA initiator	● Phase 2
FR-036	When Requestor adds external fleet equipment to Request, he must provide "Justification"	● Phase 2
FR-037	When Requestor adds external fleet equipment, "Cost Center" must be filled; prefilled from PSWS	● Phase 2
FR-038	Each equipment item added to Request is treated as a separate booking	✓ Confirmed
FR-039	Each booking has its own unique booking ID, start and end datetimes	✓ Confirmed
FR-040	System validates availability of equipment before booking	✓ Confirmed
FR-041	Equipment attributes (picture, fleet, description, usage status, etc.) are displayed	✓ Confirmed

ID	Requirement	Status
FR-042	Each booking has its own lifecycle (Draft, Submitted, Confirmed, Declined, Revoked, Terminated, Completed)	✓ Confirmed
 FR-NEW-18	Booking start datetime represents the start of mobilization , not arrival at work site (<i>Начало бронирования = начало мобилизации, не прибытие на объект</i>)	✓ Confirmed
 FR-NEW-19	No mandatory buffer between bookings; equipment availability is validated for the requested period only (<i>Нет обязательного буфера между бронями</i>)	✓ Confirmed
 FR-NEW-20	When Requestor adds Unwheeled (Not self-propelled) equipment, the system displays a notification: "Transportation equipment will be added to this booking type" (<i>При добавлении Unwheeled техники — уведомление о добавлении техники для перемещения</i>)	● Preliminary

6.6 Approval Flow (Internal Fleet Equipment)

ID	Requirement	Status
FR-043	System automatically assigns approver based on fleet ownership	✓ Confirmed
FR-044	If equipment belongs to internal fleet, booking can be confirmed or declined by Fleet Owner (member of corresponding AAD group)	✓ Confirmed
FR-045	Fleet owner can confirm incoming bookings of equipment in his fleet	✓ Confirmed
FR-046	Fleet owner can replace the equipment in the booking with another equipment before or after confirmation if booking date is not started or booking end date is in future	✓ Confirmed
FR-047	Fleet owner cannot replace the equipment if booking is revoked, terminated or booking end date is in past	✓ Confirmed

ID	Requirement	Status
FR-048	Internal fleet owner can adjust booking date range before confirmation ; equipment availability in DB is updated	 Confirmed  See CONFLICT-01 below
FR-049	Fleet owner can decline incoming bookings of equipment in his fleet; equipment availability in DB is updated	 Confirmed
FR-050	Fleet owner can indicate 'reason/comment' in case of declining	 Confirmed
 FR-NEW-21	FO must respond to a booking request within a configurable timeout period. Default: 24h — reminder notification; 48h — escalation to manager. Timeout is tracked per RequestItem (not per whole request) (<i>Тайм-аут ответа FO — 24ч напоминание, 48ч эскалация; отсчёт на уровне позиции заявки</i>)	 Confirmed
 FR-NEW-22	If FO has free equipment of the requested type but declines, FO must provide a mandatory reason (<i>При отказе при наличии свободной техники — FO обязан указать причину</i>)	 Confirmed
 FR-NEW-23	Reminder notification is sent only to the FO of the specific selected equipment item (not to all FOs who could manage that equipment type) (<i>Reminder отправляется только FO выбранной единицы техники</i>)	 Confirmed (OQ-24 closed)
 FR-NEW-24	FO presses "Mobilization started" button to record the actual mobilization start time in the system (<i>Кнопка «Мобилизация началась» у FO — фиксирует старт мобилизации</i>)	 Confirmed

CONFLICT-01 (FR-048 vs. OQ-40): BRD states FO can adjust booking period *before confirmation* — implying it is NOT allowed after confirmation. Meeting discussions (09.04) raised the question of whether FO should be able to edit the period after confirmation. **Decision pending** — OQ-40 open with TCO. Until resolved, FR-048 applies as written.

6.7 Approval Flow (External Fleet Equipment)

! MAJOR SCOPE CHANGE vs. original BRD (08.04):

Original BRD described a full two-step approval flow for on-demand BP equipment: CC Owner → External FO → DCT RWA draft creation. **Meeting decision (08.04):** On-demand BP equipment is **NOT bookable through the system in Phase 1**. It is displayed as a showcase/catalog only. All FR-051..057 are deferred to **Phase 2**.

ID	Requirement	Status
FR-051	External fleet booking requires CC owner confirmation	 Phase 2
FR-052	System identifies CC owner via PSWS integration	 Phase 2
FR-053	System allows DOAs of CC owners to confirm/decline bookings	 Phase 2
FR-054	System identifies if user is DOA of CC owner via PSWS and DOA integration	 Phase 2
FR-055	CC owner/DOA can put 'reason/comment' when declining	 Phase 2
FR-056	After CC owner confirmation, system requires External FO confirmation	 Phase 2
FR-057	External FO can confirm or decline the booking	 Phase 2
 FR-NEW-25	BP showcase (catalog): Requestor can view on-demand BP equipment by type; each entry shows contract owner and availability condition; no booking action available (<i>Витрина BP: просмотр без бронирования; показывает contract owner и условие</i>)	 Phase 1
 FR-NEW-26	BP equipment prices/rates are NOT displayed in the system (<i>Ставки BP не отображаются в системе</i>)	 Confirmed
 FR-NEW-27	BP populates their own equipment cards in the catalog; contract owner is indicated per entry (<i>BP самостоятельно заполняет карточки техники в каталоге</i>)	 Confirmed
 FR-NEW-	After FO response timeout with no available internal equipment, Requestor sees a button "Go to BP catalog" in the request (<i>После</i>	 Confirmed

ID	Requirement	Status
28	<i>тайм-аута FO при отсутствии свободной внутренней техники — кнопка «Перейти в каталог ВР»)</i>	

6.8 Booking Lifecycle Management

ID	Requirement	Status
FR-058	Requestor/SWP can revoke (withdraw) booking if not yet processed by FO; equipment availability updated	✅ Confirmed
FR-059	FO of internal fleet can terminate already confirmed booking; can indicate reason; equipment availability updated	✅ Confirmed
FR-060	Requestor/SWP can terminate already confirmed booking; equipment availability updated	✅ Confirmed (internal fleet); 🟦 Phase 2 for external
FR-061	Requestor/SWP can extend previously confirmed booking; equipment availability updated	✅ Confirmed
 FR-NEW-29	Three-step flow for Unwheeled (Not self-propelled) equipment: <i>(Трёхшаговый флоу для Unwheeled техники)</i> (1) Requestor creates request → FO; (2) FO adds transportation equipment → system sends request to "Transportation Responsible" role; (3) Transportation Responsible confirms/declines transportation readiness; (4) FO receives confirmation and gives final approval (FO can only confirm if transportation is confirmed)	✅ Confirmed (OQ-41 for UI details)
 FR-NEW-30	If transportation equipment is unavailable for Unwheeled request, FO proposes the nearest available date <i>(Если техника для перемещения недоступна — FO предлагает ближайшую дату)</i>	⚠️ Decision pending
 FR-NEW-31	Booking can be closed early by Requestor or FO via "Close early" button <i>(Досрочное закрытие — кнопка «Close early» у Requestor или FO)</i>	⚠️ Clarification pending

6.9 Booking/Request Status Management

! CONFLICT-02 (FR-070/FR-071 vs. R-75):

BRD (FR-070): "Confirmed booking status will change to 'In progress' once system datetime $\geq$ booking datetime" (automatic transition)

BRD (FR-071): "Confirmed booking status will change to 'Completed' once system datetime $<$ booking datetime" (automatic transition)

Meeting decision (09.04, R-75): Booking/request closure is **manual only** — pressed by Requestor or FO. Auto-close by scheduled time does NOT apply (equipment may be working beyond planned time).

Resolution needed: Explicit sign-off required from TCO on whether FR-070/FR-071 auto-transitions are implemented, or replaced by manual close (R-75). Currently treating R-75 as the accepted decision.

ID	Requirement	Status
FR-062	Booking will have 'Submitted' status once Request is submitted	✓ Confirmed
FR-063	Booking will have 'Confirmed' status once confirmed by FO	✓ Confirmed
FR-064	Booking will have 'Declined' status once declined by FO	✓ Confirmed
FR-065	Booking of external fleet will have 'Partially Confirmed' status once confirmed by CC Owner	● Phase 2
FR-066	Booking of external fleet will have 'Declined' status once declined by CC Owner	● Phase 2
FR-067	Status of booking returns to 'Submitted' once extended by Requestor/SWP	✓ Confirmed
FR-068	Booking will have 'Revoked' status once revoked by Requestor/SWP	✓ Confirmed
FR-069	Booking will have 'Terminated' status once terminated by Requestor/SWP/FO	✓ Confirmed

ID	Requirement	Status
FR-070	Confirmed booking status changes to 'In Progress' once system datetime >= booking datetime	! Conflict — see CONFLICT-02 above
FR-071	Confirmed booking status changes to 'Completed' once system datetime < booking datetime	! Conflict — see CONFLICT-02 above
FR-072	Request will have 'Draft' status once saved as draft by Requestor/SWP	✓ Confirmed
FR-073	Service Work Request will have 'Draft' status once imported from JDE E1	✓ Confirmed
FR-074	Request will have 'In Progress' status once at least one Booking within the Request has status 'In Progress'	⚠ Partially — aggregated statuses TBD (OQ-36)
FR-075	Request will have 'Completed' status once system datetime >= maximum end datetime	! Conflict — see CONFLICT-02; manual close (R-75) may supersede this
 FR-NEW-32	Booking closure is manual only: Requestor or FO presses "Close" button to complete a booking/request. Automatic closure by scheduled end time does NOT apply (<i>Закрытие только по кнопке; автозакрытие по времени не применяется</i>)	✓ Confirmed — supersedes FR-070/FR-071/FR-075 until sign-off
 FR-NEW-33	At the time of closing, Requestor inputs actual start time and actual end time of work; the difference between booking time and actual time represents mobilization time; data is used for utilization analytics (<i>При закрытии Requestor вводит фактическое время начала и окончания работ</i>)	✓ Confirmed (OQ-41 for UI fields)

6.10 Notifications

Note: Notifications block (FR-076..090) has **near-zero coverage in requirements registry** as of v5. The requirements below are from the original BRD and have not been explicitly confirmed or modified in meetings. Phase 1 applicability is noted based on Phase 1 scope decisions.

ID	Requirement	Status
FR-076	System notifies SWP(s) once Service Work Request is created automatically via API (Call to action)	✅ (not discussed, presumed confirmed)
FR-077	System notifies FO once Requestor/SWP submits a Request with internal fleet equipment (Call to action)	✅ Confirmed
FR-078	System notifies FO once Requestor/SWP revokes booking of internal fleet equipment (Info)	✅ (not discussed, presumed confirmed)
FR-079	System notifies Requestor/SWP about results once FO confirms or declines internal fleet booking (Success/Fail)	✅ (not discussed, presumed confirmed)
FR-080	System notifies Requestor/SWP once FO terminates a confirmed internal fleet booking (Fail result)	✅ (not discussed, presumed confirmed)
FR-081	System notifies FO once Requestor/SWP wants to extend internal fleet booking (Call to action)	✅ (not discussed, presumed confirmed)
FR-082	System notifies CC owner(s) once Requestor/SWP submits a Request with external fleet equipment (Call to action)	🟦 Phase 2
FR-083	System notifies CC owner once Requestor/SWP revokes external fleet booking before CC confirmation (Info)	🟦 Phase 2
FR-084	System notifies Requestor/SWP once CC owner declines external fleet booking (Fail result)	🟦 Phase 2
FR-085	System notifies External FO once CC owner confirms external fleet booking (Call to action)	🟦 Phase 2

ID	Requirement	Status
FR-086	System notifies CC owner and FO once Requestor/SWP revokes external fleet booking after CC confirmation but before FO confirmation (Info)	 Phase 2
FR-087	System notifies Requestor/SWP once External FO declines external fleet booking (Fail result)	 Phase 2
FR-088	System notifies Requestor/SWP once External FO confirms external fleet booking (Success + Call to action to submit RWA in DCT)	 Phase 2
FR-089	System notifies External FO once Requestor/SWP terminates confirmed external fleet booking (Info)	 Phase 2
FR-090	System notifies CC owner once Requestor/SWP wants to extend confirmed external fleet booking (Call to action)	 Phase 2
 FR-NEW-34	System sends reminder notification to FO of specific equipment after 24h of no response; escalation notification to FO's manager after 48h (<i>Напоминание FO через 24ч; эскалация на руководителя через 48ч</i>)	 Confirmed
 FR-NEW-35	System notifies Transportation Responsible role when FO adds transportation equipment to an Unwheeled booking request (<i>Уведомление роли «Транспортировка» при добавлении заявки на перемещение</i>)	 Confirmed
 FR-NEW-36	System notifies FO when Transportation Responsible confirms or declines transportation readiness for Unwheeled booking (<i>Уведомление FO о решении роли «Транспортировка»</i>)	 Confirmed

6.11 Search & Reporting

ID	Requirement	Status
FR-091	Requestor / SWP can search and filter own requests	 Confirmed
FR-092	Approver (CC Owner / FO) can view all pending and completed approvals	 Confirmed

ID	Requirement	Status
FR-093	Admin can view all requests	✓ Confirmed
FR-094	System generates reports on Requests / Bookings / Equipments with capability to apply various filters	✓ Confirmed
FR-095	Admin can view/download reports on all Requests / Bookings / Equipments	✓ Confirmed
FR-096	Approver (CC Owner / FO) can view/download reports on pending and completed Requests / Bookings	✓ Confirmed
FR-097	FO can view/download reports on own equipments	✓ Confirmed
FR-098	SWPs (and selected FOs — Maintenance) can view Service Work Requests grouped in Work Orders	✓ Confirmed
NEW FR-NEW-37	Audit trail and history of all bookings available on demand (not daily/weekly scheduled reports) <i>(Аудит и история бронирований по запросу, не периодические отчёты)</i>	✓ Confirmed
NEW FR-NEW-38	Search results for equipment must display as mandatory fields: state registration number + TCO equipment number + model/make <i>(В результатах поиска обязательно: госномер + ТСО-номер + модель/марка)</i>	✓ Confirmed
NEW FR-NEW-39	Search filters in equipment search are dynamic : only parameters relevant to the selected equipment type are displayed; filter matrix (equipment type → filter set) to be agreed with TCO <i>(Динамические фильтры поиска по типу техники; матрица «тип → фильтры» согласуется с ТСО)</i>	⚠ Pending (OQ-29)

6.12 Dashboards / Map

ID	Requirement	Status
FR-099	System has deployed/integrated visualization tool displaying utilization and coordinates from DataLake/PI (tracker data)	✓ Confirmed

ID	Requirement	Status
FR-100	System has deployed/integrated visualization tool displaying digitized location of Work Orders from DataLake (JDE E1)	 Confirmed

6.13 Audit & Compliance

ID	Requirement	Status
FR-101	System logs all request creation, modifications, approvals, rejections	 Confirmed
FR-102	Audit trail accessible for compliance purposes	 Confirmed
FR-103	Exportable logs for legal/regulatory audits	 Confirmed

7 Non-Functional Requirements

Area	Requirement	Status
Availability	99.5% uptime	 Confirmed
Performance	The system must support high volume of requests per hour without performance degradation	 Confirmed
Security	Role-based access control (RBAC); system integrates with corporate authentication (SSO / Azure AD)	 Confirmed
Compliance	Maintain audit history of all approvals / rejections / revokes / terminations	 Confirmed
Scalability	Support both internal and external partner use cases	 Confirmed
Usability	Intuitive UI for both requesters and approvers	 Confirmed
 Users	~200 users (per ACR assessment, 07.04); No PII data stored	 Confirmed

Area	Requirement	Status
 Security tier	ACR assessment result: Low Risk (07.04)	 Confirmed
 Authentication	Azure SSO for internal users; separate account flow for external BP users (Phase 2)	 Confirmed
 UX reference	UI simplicity reference: Booking.com (<i>Ориентир по простоте UI: Booking.com</i>)	 Confirmed

8 Integrations

8.1 JDE E1

Work Orders created in JDE E1 must be imported to HDV/HDE tool as Service Work Request(s), once a certain (TBD) status is assigned to the Work Order. Work Order can contain several Steps; each step is created as a separate Service Work Request in HDV/HDE tool. Only Steps with certain types (BHOE, CDECK, DTRK, FORKL, FTRK-ХИНО, GDRV, GMLOG, MLIFT, VTRK, etc.) will be imported.

The systems must have backward integration which allows changing the status of 'Step' in JDE based on status change of the imported 'Request' in HDV/HDE or by Service Work Processor's manual actions.

Status:  Partially confirmed — JDE trigger status TBD (R-30)

 **Additional (08.04):** Repair status (maintenance in progress / planned completion date) is sourced from JDE SQL database and displayed in Requestor's equipment search. —  Details TBD (OQ — JDE integration)

8.2 DCT (Digital Contractor Timesheet)

8.2.1 RWA / Change Order Creation

When external fleet equipment booking receives all approvals, system creates draft of RWA in DCT (via POST API). When booking prolongation receives all approvals, system creates Change Order in DCT.

Status:  Phase 2 (external fleet booking workflow is Phase 2)

8.2.2 Tracker's data receival

Once external fleet equipments are equipped with trackers, system sends tracker data to DCT for actual vs. reported hours comparison.

Status:  Phase 2

8.3 PSWS (TCO White Page)

Identifies CC owner of cost centers for external fleet booking approval; provides preferred email address for notification recipients.

Status:  Phase 2 (CC Owner role is Phase 2)

8.4 DOA (Delegation of Authority)

Identifies if user is DOA of CC owner for external fleet booking processing.

Status:  Phase 2

8.5 DataLake / PI

From DataLake: utilization information from IVMS and WIALON trackers (extendable to other trackers). Equipment "Tracker ID" field allows system to merge equipment records with DataLake utilization and location data.

Status:  Confirmed

Note (07.04): WIALON integration is **blocked** pending cyber security assessment (Wialon is currently on the restricted list). IVMS and Vega trackers are in scope. — 

 **GIS / Map integration (07.04):** Integration with MAPH/Atlas GIS tool via iframe + URL parameters; equipment ID synchronization; Azure SSO pass-through. —  Pending cyber assessment approval (Светлана)

9 Open Conflicts Summary

#	Conflict	BRD Position	Meeting Decision	Action Required
CONFLICT-01	FO editing booking period after confirmation	FR-048: editing allowed "before confirmation" (implies not after)	OQ-40: open question — can FO edit after confirmation?	Sign-off from TCO
CONFLICT-02	Booking status auto-transition by datetime	FR-070/071/075: automatic status change by datetime	R-75 (09.04): manual close only	Sign-off from TCO; update state machine
CONFLICT-03	External fleet booking workflow	FR-051..057: full CC Owner → External FO flow	08.04: on-demand BP = showcase only in Phase 1; full flow is Phase 2	Explicitly document Phase 1/2 split in project scope

10 Requirements Not in Original BRD (New from Meetings)

All requirements below were identified during requirements gathering sessions and are **not present in original BRD**. They require formal acknowledgement from TCO.

New ID	Summary	Source	Status
FR-NEW-01	All time parameters configurable via Admin Panel	03.04	✓
FR-NEW-02	FO delegation to any TCO employee with auto-revoke	08.04	✓
FR-NEW-03	Admin configures FO timeout thresholds	03.04	✓

New ID	Summary	Source	Status
FR-NEW-04	Admin authorizes users to book Assigned equipment	08.04	✓
FR-NEW-05	FO uploads photos; Requestor sees photo	08.04	● Preliminary
FR-NEW-06	Repair status from JDE displayed in search	08.04	⚠
FR-NEW-07	Stationary HDE excluded from Requestor search	08.04	✓
FR-NEW-08	Assigned equipment: visible to all, bookable by authorized only	08.04	✓
FR-NEW-09	Shared with Conditions has visual color marker	03.04	✓
FR-NEW-10	On-demand only; no weekly schedules	03.04	✓
FR-NEW-11	Priority P1–P4 selected by Requestor	03.04	✓
FR-NEW-12	Work Order mandatory; Location replaces it for SCM	08.04	✓
FR-NEW-13	"Work Description" mandatory (~50 chars)	08.04	✓
FR-NEW-14	"Comments" optional	08.04	✓
FR-NEW-15	Single form → auto-split into atomic bookings via WO	09.04	✓
FR-NEW-16	Partial confirmation per RequestItem	03.04	✓
FR-NEW-17	Aggregated Request status auto-calculated	09.04 arch	⚠
FR-NEW-18	Booking start = mobilization start	03.04	✓
FR-NEW-19	No mandatory buffer between bookings	09.04	✓
FR-NEW-20	Unwheeled equipment warning notification in UI	07.04	●

New ID	Summary	Source	Status
FR-NEW-21	FO timeout: 24h reminder / 48h escalation per RequestItem	03.04, 06.04	✓
FR-NEW-22	FO must state reason for decline if equipment available	06.04	✓
FR-NEW-23	Reminder to FO of specific equipment only	07.04	✓
FR-NEW-24	FO "Mobilization started" button	09.04	✓
FR-NEW-25	BP showcase: view-only catalog by type	08.04	✓
FR-NEW-26	BP prices not displayed	08.04	✓
FR-NEW-27	BP self-populates catalog cards	08.04	✓
FR-NEW-28	"Go to BP catalog" button after FO timeout	08.04	✓
FR-NEW-29	Three-step Unwheeled flow with Transportation role	09.04	✓
FR-NEW-30	FO proposes nearest date if transport unavailable	03.04	⚠
FR-NEW-31	Early close button for Requestor/FO	07.04	⚠
FR-NEW-32	Booking closure is manual only (overrides auto-datetime)	09.04	✓ ! Conflict
FR-NEW-33	Actual start/end time entered at closure	09.04	✓
FR-NEW-34	Reminder to FO (24h) + escalation to manager (48h)	03.04	✓
FR-NEW-35	Notify Transportation Responsible on Unwheeled request	09.04	✓
FR-NEW-36	Notify FO on Transportation decision	09.04	✓
FR-NEW-37	Audit/history on demand, not scheduled	09.04	✓

New ID	Summary	Source	Status
FR-NEW-38	Search results: state reg. number + TCO number + model	08.04	✓
FR-NEW-39	Dynamic filters per equipment type (matrix TBD)	08.04	⚠
Transportation role (4.6)	Dedicated role for Heavy Ops / Maintenance transport team	09.04	✓
Equipment snapshot (5.2.3)	Snapshot of equipment data at booking confirmation	09.04 arch	⚠